

ENTERPRISE BUSINESS REQUIREMENTS DOCUMENTATION

AI-Powered Healthcare Medical Institution Platform

Business Requirements Document

Volume I of IV | Business Vision, Objectives, Scope & Functional Requirements

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1.0	September 2026	Enterprise Architecture & Digital Health Strategy	Initial draft issued for executive review.

1. Project Vision

The AI-Powered Healthcare Medical Institution Web Application is envisioned as a unified digital platform that connects patient care, clinical operations, hospital administration, and healthcare analytics inside a single, secure, AI-augmented environment. The platform is designed to support the full patient journey — from registration, check-in, and diagnosis through monitoring, medication management, admission, discharge, and checkout — while giving clinical staff and administrators real-time visibility into operational and health status information.

Central to the vision is the responsible integration of artificial intelligence: a governed, customized large language model capability supports AI-assisted healthcare education, AI-assisted reporting, narration, and multimedia generation (text-to-image and text-to-audio), all operating under a defined Responsible AI governance framework rather than as an unconstrained, general-purpose assistant. AI features are positioned as decision-support and productivity aids for clinical and administrative staff and as educational aids for patients — not as an autonomous clinical decision-maker.

1.1 Business Value

- Consolidates patient, clinical, and administrative data into a single system of record, reducing duplicate entry and fragmented workflows across departments.
- Improves institutional efficiency by digitizing check-in/checkout, admission/discharge, and medication tracking workflows currently handled manually or across disconnected tools.
- Creates a foundation for data-driven decision-making through healthcare operational analytics and executive dashboards.

1.2 Healthcare Value

- Supports safer patient care through consolidated diagnosis history, medication records, and severity/monitoring indicators visible to authorized clinical staff.
- Reduces time-to-information for physicians and nurses during time-sensitive emergency room and inpatient scenarios.
- Enables structured, auditable tracking of a patient's clinical journey across admission, treatment, and discharge.

1.3 Administrative Value

- Provides administrators and co-administrators with controlled tools for account management, role assignment, and audit oversight.
- Standardizes administrative approval workflows for sensitive actions (e.g., record changes, access-level changes, discharge approvals).
- Reduces administrative overhead through centralized CRUD operations for patient and administrative records.

1.4 Operational & Analytical Value

- Delivers real-time and historical visibility into emergency room load, admissions, discharges, and medication activity.
- Surfaces patient severity and health-score metrics to support triage and resource allocation decisions.
- Provides configurable dashboards for executives, analysts, and department leads.

1.5 Technological Value

- Establishes an enterprise-grade, scalable architecture suitable for phased rollout, beginning with a conceptual build on the Replit AI Website Builder platform.
- Introduces a governed, customizable LLM layer (supported by Claude AI in the requirements and architecture phase) rather than ad hoc AI tooling.
- Builds in interoperability considerations for future integration with external clinical and administrative systems.

1.6 Educational Value

- AI-generated healthcare educational content helps patients understand diagnoses, procedures, and medication instructions in accessible language.
- AI narration and text-to-audio capabilities improve accessibility for patients with visual impairment or low literacy.
- Text-to-image generation supports illustrative, non-diagnostic visual aids for patient education (e.g., anatomy or process illustrations), clearly distinguished from diagnostic imaging.

2. Business Objectives

This section defines measurable objectives across fourteen business domains, and distinguishes objectives, outcomes, benefits, success criteria, and KPIs so each term is used consistently throughout the document.

2.1 Definitions Used in This Document

TERM	DEFINITION
Business Objective	A specific, intended result the platform is built to achieve.
Business Outcome	The observable change in operations or care delivery once the objective is met.
Business Benefit	The value realized by the institution or its patients as a result of the outcome.
Success Criteria	The condition(s) that must be true to consider the objective achieved.
Key Performance Indicator (KPI)	The specific metric tracked over time to measure progress toward the objective.

2.2 Patient Management

Business Objective

Provide a single, accurate, and complete digital record for each patient across registration, visits, and care episodes.

Business Outcome

Clinical and administrative staff access one authoritative patient record instead of fragmented paper or siloed systems.

Business Benefit

Reduced duplicate data entry, fewer record-matching errors, faster patient lookup.

Success Criteria

100% of active patients have a single digital profile; no duplicate-record rate above defined threshold.

Key Performance Indicator(s)

Duplicate patient record rate; average patient lookup time.

2.3 Healthcare Operations**Business Objective**

Digitize and standardize check-in, admission, discharge, and emergency room workflows.

Business Outcome

Staff execute operational workflows through the platform rather than manual/paper processes.

Business Benefit

Reduced patient wait times and operational bottlenecks.

Success Criteria

Defined percentage of check-in/admission/discharge events processed digitally within target timeframes.

Key Performance Indicator(s)

Average check-in time; average admission-to-bed-assignment time; ER throughput rate.

2.4 Administrative Efficiency**Business Objective**

Centralize administrative account management, role assignment, and approval workflows.

Business Outcome

Administrators manage accounts and permissions through a governed, auditable interface.

Business Benefit

Reduced administrative processing time and reduced unauthorized access incidents.

Success Criteria

All administrative actions logged and attributable; approval workflow completion within SLA.

Key Performance Indicator(s)

Average administrative task completion time; percentage of actions with complete audit trail.

2.5 Clinical Information Organization**Business Objective**

Structure diagnosis, medication, and history data in a consistent, retrievable format.

Business Outcome

Clinicians retrieve relevant clinical history without cross-referencing multiple systems.

Business Benefit

Improved continuity of care and reduced clinical documentation time.

Success Criteria

Defined percentage of clinical records structured per data model standards.

Key Performance Indicator(s)

Record completeness rate; clinician time spent per documentation task.

2.6 Healthcare Analytics

Business Objective

Provide operational and clinical metrics (severity, admissions, discharges, medication, ER load) through dashboards.

Business Outcome

Decision-makers access near-real-time operational metrics rather than retrospective manual reports.

Business Benefit

Faster, better-informed operational and resource decisions.

Success Criteria

Core metrics refreshed within defined latency; dashboards adopted by target user groups.

Key Performance Indicator(s)

Dashboard usage rate; metric refresh latency; decision cycle time.

2.7 AI-Assisted Healthcare Education

Business Objective

Generate accurate, accessible educational content for patients regarding diagnoses, procedures, and medications.

Business Outcome

Patients receive supplementary educational material aligned with their care plan.

Business Benefit

Improved patient understanding and engagement with care instructions.

Success Criteria

Educational content reviewed/approved per governance process before patient delivery.

Key Performance Indicator(s)

Patient engagement rate with educational content; content review compliance rate.

2.8 AI-Assisted Reporting

Business Objective

Use governed AI capabilities to accelerate drafting of operational and administrative reports.

Business Outcome

Staff use AI-assisted drafts as a starting point, subject to human review, rather than manual report authoring from scratch.

Business Benefit

Reduced report preparation time; more consistent reporting format.

Success Criteria

All AI-assisted reports pass through defined human review before distribution.

Key Performance Indicator(s)

Report preparation time reduction; human-review completion rate.

2.9 Patient Monitoring

Business Objective

Provide structured, near-real-time visibility into patient status and severity indicators to authorized staff.

Business Outcome

Clinical staff receive timely status/severity information to support triage and prioritization.

Business Benefit

Improved responsiveness to changes in patient condition.

Success Criteria

Monitoring data refreshed within defined latency for in-scope patient categories.

Key Performance Indicator(s)

Monitoring data refresh latency; staff response time to flagged severity changes.

2.10 Data Governance

Business Objective

Establish clear ownership, access control, and lifecycle rules for patient and administrative data.

Business Outcome

Data access, retention, and modification follow documented governance policy.

Business Benefit

Reduced data-related compliance risk and improved data trustworthiness.

Success Criteria

Governance policy documented, approved, and enforced through access controls.

Key Performance Indicator(s)

Policy compliance audit pass rate; unauthorized access incident count.

2.11 Responsible AI

Business Objective

Ensure AI features operate within a documented Responsible AI governance framework covering transparency, human oversight, and bias/quality review.

Business Outcome

AI-generated content and AI-assisted recommendations are reviewed and clearly labeled as AI-assisted.

Business Benefit

Reduced risk of AI-related clinical or administrative errors and increased staff/patient trust.

Success Criteria

All AI-generated patient-facing or clinical-support content passes governance review before use.

Key Performance Indicator(s)

Percentage of AI outputs labeled and reviewed per policy; AI-related incident count.

2.12 Security and Privacy

Business Objective

Protect PII/PHI through access controls, encryption, and auditability aligned with applicable healthcare data protection requirements.

Business Outcome

Patient and administrative data are accessed only by authorized roles, with a complete audit trail.

Business Benefit

Reduced risk of data breach and improved regulatory posture.

Success Criteria

Zero unresolved critical security findings at go-live; full audit logging in place.

Key Performance Indicator(s)

Number of access-control violations; audit log completeness rate.

2.13 Scalability

Business Objective

Design the platform to scale from an initial single-institution deployment to a multi-facility footprint.

Business Outcome

The system supports growth in patient volume, user accounts, and data volume without redesign.

Business Benefit

Reduced future re-platforming cost and risk.

Success Criteria

Architecture documented to support defined future growth scenarios.

Key Performance Indicator(s)

System performance under projected load; scalability test pass rate.

2.14 Usability

Business Objective

Provide an intuitive, role-appropriate interface for clinical, administrative, and patient-facing users.

Business Outcome

Users complete core workflows without extensive training.

Business Benefit

Faster adoption and reduced user error.

Success Criteria

Defined task-completion rate and satisfaction score achieved in usability testing.

Key Performance Indicator(s)

Task success rate; user satisfaction score; training time required.

2.15 Digital Transformation

Business Objective

Shift the institution from manual/paper-based and siloed digital processes to a unified, AI-augmented digital platform.

Business Outcome

Core patient, clinical, and administrative workflows operate primarily through the platform.

Business Benefit

Improved organizational agility, data visibility, and long-term operating efficiency.

Success Criteria

Defined percentage of target workflows migrated to the platform within the implementation roadmap.

Key Performance Indicator(s)

Workflow digitization rate; legacy-process retirement rate.

3. Project Scope

3.1 In-Scope

- Patient registration, profile management, and identification (PII/PHI handling under governed access control).
- Patient check-in and checkout workflows.
- Hospital admission and discharge workflows.
- Medical diagnosis information capture and retrieval.
- Medication management and tracking.
- Emergency room operations tracking.
- Patient monitoring and severity indicators.
- Administrative account management (administrators and co-administrators), role assignment, and permission management.
- Administrative approval workflows and audit logging.
- Healthcare operational analytics and dashboards (severity, admissions, discharges, ER, medication, check-in metrics).
- AI-assisted healthcare educational content generation (subject to Responsible AI review).
- AI-assisted reporting (draft generation subject to human review).
- Text-to-image and text-to-audio generation for educational/accessibility purposes (non-diagnostic).
- AI narration capability for educational content and accessibility.
- Responsible AI governance framework and documentation.
- Core public-facing pages (Home, About Us, Contact Us, Services) and secured pages (Patient Data, Dashboard, Admin Login, AI Features, Reporting, Admin Portal, Admin Dashboard).

3.2 Out-of-Scope (Initial Implementation)

- Direct integration with third-party Electronic Health Record (EHR) or Electronic Medical Record (EMR) systems.

- Autonomous AI clinical diagnosis or treatment decision-making (AI remains decision-support/educational only).
- Insurance claims processing and billing/payment functionality.
- Telemedicine/video consultation capabilities.
- Native mobile applications (initial scope is a responsive web application).
- Integration with medical device telemetry/IoT monitoring hardware.
- Multi-facility / multi-tenant deployment (planned for future phase).
- Pharmacy inventory and supply-chain management.
- Real-time bed/resource optimization algorithms.

3.3 Future Scope

- Integration with external EHR/EMR and health information exchange (HIE) systems.
- Multi-facility and multi-tenant architecture for healthcare networks.
- Native mobile applications for patients and clinical staff.
- Integration with wearable/IoT patient monitoring devices for real-time vitals.
- Expanded AI capabilities such as predictive severity modeling (with appropriate clinical validation and governance).
- Telemedicine and virtual consultation modules.
- Insurance and billing integration.
- Advanced predictive analytics for capacity planning and resource optimization.
- Expanded accessibility and multilingual AI narration/education support.

4. Stakeholder Requirements

The following matrix identifies primary stakeholder groups, their responsibilities within the platform, the system capabilities they require, their access level, and their primary goals.

STAKEHOLDER	RESPONSIBILITIES	SYSTEM NEEDS	ACCESS LEVEL	PRIMARY GOALS
Patients	Manage own profile; view own diagnosis/medication/history; complete check-in.	Patient portal, educational AI content, accessible narration.	Self-service (own record only)	Understand and manage their own care journey.
Physicians	Record diagnoses; review patient history; monitor severity; guide treatment.	Patient clinical record access, monitoring dashboards, AI-assisted reporting.	Clinical (assigned patients)	Deliver accurate, timely, well-informed care.
Nurses	Check-in/monitoring support; medication administration tracking; escalation.	Patient monitoring views, medication tracking tools.	Clinical (assigned patients/unit)	Maintain accurate real-time patient status.
Clinical Staff	Support diagnostic and treatment workflows; document care activities.	Shared clinical record access, task/workflow tools.	Clinical (role-restricted)	Efficient, accurate clinical documentation.
Hospital Administrators	Manage patient/administrative CRUD; approve workflows; oversee	Admin portal, approval workflows, operational dashboards.	Administrative (institution-wide)	Efficient, compliant institutional operations.

STAKEHOLDER	RESPONSIBILITIES	SYSTEM NEEDS	ACCESS LEVEL	PRIMARY GOALS
	operations.			
System Administrators	Manage accounts, roles, permissions, system configuration.	Admin dashboard, role/permission management, audit logs.	System (full technical access)	Secure, reliable, well-governed system operation.
Healthcare Executives	Review operational and strategic metrics; guide resource decisions.	Executive dashboards, analytics reporting.	Analytics (aggregate/read-only)	Data-driven strategic and operational decisions.
Data Analysts	Analyze operational/clinical metrics; build reports.	Analytics tools, reporting exports, dashboard configuration.	Analytics (aggregate, de-identified where applicable)	Accurate, actionable operational insight.
Compliance Personnel	Monitor regulatory and policy adherence; review audit trails.	Audit logs, governance documentation, compliance reporting.	Compliance (audit/read access)	Verifiable regulatory and policy compliance.
Responsible AI Governance Personnel	Review AI-generated content; enforce AI governance policy; manage model configuration boundaries.	AI review queues, governance dashboards, content approval tools.	AI Governance (AI outputs and configuration)	Safe, transparent, well-governed AI usage.
Authorized Healthcare Personnel	Access relevant patient information within role boundaries for care coordination.	Role-scoped patient record access.	Clinical/ Administrative (role-scoped)	Coordinated, informed patient care.
Technical Support Personnel	Provide user support; troubleshoot system issues; escalate technical defects.	Support tools, system status dashboards, ticketing integration.	Technical (system diagnostics, non-clinical data)	Reliable system availability and user support.

5. Functional Requirements

Functional requirements are organized by domain and documented using a consistent structure: Requirement ID, Requirement, Description, Business Purpose, Priority, Primary User, and Acceptance Criteria. Priority uses the scale: Must Have, Should Have, Could Have (MoSCoW).

5.1 Patient Management

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-PAT-001	Patient Registration	Capture new patient demographic, contact, and identification information.	Establish accurate patient identity as system of record.	Must Have	Admin / Clinical Staff	New patient profile created and retrievable within the system.
FR-PAT-002	Patient Profile Management	Create, view, update patient profile including PII and next-of-kin details.	Maintain accurate, current patient information.	Must Have	Admin / Clinical Staff	Profile updates saved with audit trail and timestamp.
FR-PAT-003	Patient Identification (PII)	Securely store and retrieve personally identifiable information.	Ensure accurate identity verification and record matching.	Must Have	Admin / Clinical Staff	PII fields access-controlled and encrypted at rest.
FR-PAT-	Protected Health	Store and manage PHI in	Protect sensitive	Must	Clinical Staff	PHI access restricted to

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
004	Information (PHI) Handling	compliance with governance policy.	clinical data and support regulatory compliance.	Have		authorized roles; access logged.
FR-PAT-005	Diagnosis Recording	Capture and update medical diagnosis information linked to patient record.	Maintain accurate clinical history for care continuity.	Must Have	Physicians	Diagnosis entries linked to patient record with timestamp and author.
FR-PAT-006	Patient Monitoring	Display near-real-time patient status indicators to authorized staff.	Support timely clinical response to status changes.	Must Have	Physicians / Nurses	Monitoring view reflects status within defined refresh interval.
FR-PAT-007	Patient Severity Indicators	Calculate and display severity classification per patient.	Support triage and resource prioritization.	Should Have	Physicians / Nurses	Severity indicator visible and updated per defined rules.
FR-PAT-008	Medication Information	Record and display current and historical medication data.	Reduce medication errors and support treatment continuity.	Must Have	Physicians / Nurses	Medication list accurate and time-stamped per entry.
FR-PAT-009	Patient History	Maintain chronological record of visits, diagnoses, and treatments.	Support informed clinical decision-making.	Must Have	Physicians	Historical events retrievable in chronological order.
FR-PAT-010	Check-in / Checkout	Digitally process patient arrival and departure events.	Streamline front-desk and operational workflow.	Must Have	Admin / Clinical Staff	Check-in/checkout event recorded with timestamp.
FR-PAT-011	Admission / Discharge	Process hospital admission and discharge with required approvals.	Standardize inpatient workflow and bed management.	Must Have	Clinical Staff / Admin	Admission/discharge status updated and auditable.
FR-PAT-012	Next of Kin Management	Capture and maintain next-of-kin/emergency contact information.	Support patient safety and emergency communication.	Should Have	Admin / Clinical Staff	Next-of-kin record linked and retrievable per patient.

5.2 Administrative Management

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-ADM-001	Administrator Authentication	Secure login for administrative users with role verification.	Protect administrative functions from unauthorized access.	Must Have	System Admin	Authentication enforced; failed attempts logged.
FR-ADM-002	Administrator Account Creation	Create new administrator accounts with defined role assignment.	Enable controlled onboarding of administrative staff.	Must Have	System Admin	New admin account created with assigned role and audit entry.
FR-ADM-003	Co-Administrator Creation	Create secondary/co-administrator accounts with scoped permissions.	Support delegated administrative oversight.	Should Have	System Admin	Co-admin account created with defined permission scope.
FR-ADM-004	Patient Record CRUD	Create, read, update, and delete patient records within policy limits.	Maintain accurate and current patient data.	Must Have	Admin	CRUD operations logged with user, action, and timestamp.
FR-ADM-005	Administrative Record CRUD	Manage administrative and configuration records.	Support institutional operational management.	Must Have	Admin	CRUD operations restricted by role and fully logged.

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-ADM-006	Role Assignment	Assign and modify user roles across stakeholder categories.	Enforce least-privilege access control.	Must Have	System Admin	Role changes logged and take effect immediately.
FR-ADM-007	Permission Management	Configure granular permissions per role or account.	Support precise, policy-aligned access control.	Must Have	System Admin	Permission changes validated against policy rules.
FR-ADM-008	Audit Activity Log	Record all sensitive administrative and clinical actions.	Support compliance, security review, and accountability.	Must Have	Compliance / System Admin	All in-scope actions produce a retrievable audit entry.
FR-ADM-009	Administrative Approval Workflows	Route sensitive actions (e.g., discharge, access changes) through approval.	Reduce risk of unauthorized or erroneous actions.	Should Have	Admin	Action blocked until required approval is recorded.

5.3 Healthcare Operations

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-OPS-001	Emergency Room Management	Track ER patient intake, status, and throughput.	Improve ER responsiveness and resource allocation.	Must Have	Clinical Staff	ER status view reflects current patient queue and status.
FR-OPS-002	Hospital Admission Workflow	Manage end-to-end admission process including bed assignment status.	Standardize and speed up admission processing.	Must Have	Clinical Staff / Admin	Admission workflow completes with recorded approvals.
FR-OPS-003	Hospital Discharge Workflow	Manage discharge process including required sign-offs.	Ensure safe, documented patient discharge.	Must Have	Clinical Staff / Admin	Discharge recorded with physician sign-off reference.
FR-OPS-004	Patient Monitoring (Operational)	Aggregate patient monitoring data at unit/department level.	Support operational load and staffing visibility.	Should Have	Nurses / Admin	Unit-level monitoring view reflects current patient load.
FR-OPS-005	Medication Tracking	Track medication administration events across patients.	Reduce medication errors and support accountability.	Must Have	Nurses	Medication administration recorded with timestamp and staff ID.
FR-OPS-006	Operational Reporting	Generate standard operational reports (ER, admissions, discharge, medication).	Support institutional operational review.	Should Have	Admin / Analysts	Report generated with accurate, current data.

5.4 Healthcare Analytics

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-ANA-001	Patient Severity Metrics	Aggregate and visualize severity classifications across patients.	Support triage prioritization and resource planning.	Should Have	Analysts / Physicians	Severity metrics visualized and refreshed per defined interval.
FR-ANA-002	Patient Health Score Metrics	Calculate and display aggregate/individual health score trends.	Support clinical and operational trend analysis.	Could Have	Analysts / Physicians	Health score trend chart reflects underlying data accurately.

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-ANA-003	Diagnosis Metrics	Aggregate diagnosis data by category, frequency, and trend.	Support clinical and epidemiological insight.	Should Have	Analysts	Diagnosis metrics match underlying patient records.
FR-ANA-004	Admission Metrics	Track admission volume, rate, and trends over time.	Support capacity and staffing planning.	Must Have	Admin / Analysts	Admission metrics reconcile with admission records.
FR-ANA-005	Discharge Metrics	Track discharge volume, timing, and trend data.	Support bed turnover and operational efficiency analysis.	Must Have	Admin / Analysts	Discharge metrics reconcile with discharge records.
FR-ANA-006	Check-in Metrics	Track check-in volume and average processing time.	Identify front-desk operational bottlenecks.	Should Have	Admin / Analysts	Check-in metrics reflect actual check-in event data.
FR-ANA-007	Medication Metrics	Track medication administration volume and adherence trends.	Support medication safety and operational review.	Should Have	Analysts / Clinical Staff	Medication metrics reconcile with medication tracking records.
FR-ANA-008	Emergency Room Metrics	Track ER volume, wait time, and throughput.	Support ER staffing and process improvement.	Must Have	Admin / Analysts	ER metrics reflect current and historical ER activity.
FR-ANA-009	Healthcare Operational Metrics	Provide institution-wide operational KPI dashboard.	Support executive and administrative decision-making.	Must Have	Executives / Admin	Dashboard reflects defined KPI set with correct values.
FR-ANA-010	Patient Monitoring Metrics	Aggregate monitoring data for trend and exception reporting.	Identify at-risk patients and monitoring gaps.	Should Have	Analysts / Clinical Staff	Monitoring metrics flag exceptions per defined thresholds.

5.5 AI-Enabled Capabilities

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-AI-001	AI-Generated Healthcare Educational Content	Generate patient-appropriate educational content on diagnoses, procedures, and medications.	Improve patient understanding and engagement.	Should Have	Patients	Content generated, then routed through governance review before display.
FR-AI-002	AI-Assisted Reporting	Generate draft operational/administrative report content from underlying data.	Reduce manual report preparation time.	Should Have	Admin / Analysts	Draft report requires human review/approval before distribution.
FR-AI-003	Text-to-Image Generation	Generate non-diagnostic illustrative images to support patient education.	Improve comprehension of care-related concepts.	Could Have	Patients	Generated images clearly labeled as illustrative, non-diagnostic.
FR-AI-004	Text-to-Audio Generation	Convert educational/informational text into audio format.	Improve accessibility for patients with visual impairment or low literacy.	Could Have	Patients	Audio output accurately reflects source text content.
FR-AI-005	AI Narration	Provide narrated walkthroughs of educational or informational content.	Improve accessibility and patient engagement.	Could Have	Patients	Narration available and synchronized with displayed content.

ID	REQUIREMENT	DESCRIPTION	BUSINESS PURPOSE	PRIORITY	PRIMARY USER	ACCEPTANCE CRITERIA
FR-AI-006	Customized LLM Capability	Provide a governed, domain-configured LLM for internal AI features.	Ensure AI outputs are relevant, controlled, and policy-compliant.	Must Have	AI Governance / Admin	LLM configuration documented and restricted to approved use cases.
FR-AI-007	Responsible AI Review Workflow	Route AI-generated patient-facing or clinical-support content through human review.	Reduce risk of inaccurate or inappropriate AI output reaching patients/staff.	Must Have	AI Governance Personnel	No AI-generated patient-facing content published without recorded review.
FR-AI-008	AI Output Labeling	Clearly label AI-generated content as AI-assisted throughout the platform.	Maintain transparency with patients and staff.	Must Have	All Users	All AI-generated content visibly labeled in the UI.

6. Non-Functional Requirements

Non-functional requirements define the quality attributes the platform must meet. Each is stated with a measurable target to support later validation.

ATTRIBUTE	REQUIREMENT	MEASURABLE TARGET
Security	Role-based access control (RBAC), encryption at rest and in transit, and multi-factor authentication for administrative and clinical accounts.	Zero unresolved critical/high vulnerabilities at go-live.
Privacy	PII/PHI access limited to authorized roles, with purpose-based access justification for sensitive record views.	100% of PII/PHI access events logged and attributable.
Availability	Core platform available for clinical and administrative use during defined operating windows.	Target uptime of 99.9% for production environment (excluding planned maintenance).
Performance	Responsive interaction times for core workflows (check-in, record lookup, dashboard load).	95th-percentile page/API response time under 2 seconds for core workflows.
Scalability	Architecture supports growth in concurrent users, patient volume, and data volume.	Support defined initial concurrent-user target with headroom for 3x growth without redesign.
Reliability	Consistent, correct system behavior under normal and peak load.	Error rate below defined threshold (e.g., <0.5%) for core transactional workflows.
Maintainability	Modular architecture and documentation enabling efficient updates and defect resolution.	Defined mean time to resolve (MTTR) for standard-severity defects.
Accessibility	Interfaces conform to recognized accessibility standards (e.g., WCAG 2.1 AA) including AI narration support.	Pass accessibility audit at defined conformance level.
Usability	Role-appropriate, intuitive navigation with minimal training required for core tasks.	Defined task-success rate and satisfaction score in usability testing.
Interoperability	Data structures and APIs designed to support future integration with external clinical/administrative systems.	Core data models documented per interoperability standards for future integration.
Data Integrity	Validation rules and referential integrity across patient, clinical, and administrative records.	Zero critical data-integrity defects at go-live; automated validation on key entry points.
Disaster Recovery	Defined backup, replication, and recovery procedures for	Recovery Point Objective (RPO) and

ATTRIBUTE	REQUIREMENT	MEASURABLE TARGET
	critical data.	Recovery Time Objective (RTO) defined and tested.
Business Continuity	Documented procedures for continued critical operations during system disruption.	Business continuity plan documented, reviewed, and tested at defined intervals.
Monitoring	Real-time monitoring of system health, performance, and security signals.	Critical alerts triaged within defined response-time target.
Logging	Comprehensive, tamper-evident logging of sensitive actions and system events.	100% of in-scope sensitive actions produce a retained log entry.
Auditability	Ability to reconstruct who did what, when, and to which record.	Audit trail available for 100% of PII/PHI and administrative actions.
Extensibility	Architecture supports addition of future modules (e.g., telemedicine, EHR integration) without core redesign.	Documented extension points for at least the defined future-scope items in Section 3.3.

7. Website Requirements

Each page below is documented with its purpose, users, capabilities, information displayed, security considerations, data requirements, and UX expectations. The overall site follows a modern, responsive design with a light primary theme, dark hero sections, contemporary/futuristic typography, professional readability, interactive elements, and purposeful scroll effects.

7.1 Home Page

Purpose

Introduce the institution and platform; orient visitors toward relevant next steps (patient, clinical, or administrative).

Primary Users

Public visitors, prospective patients, general staff.

Key Capabilities

Navigation to core sections; high-level overview of services and AI capabilities; call-to-action for login/registration.

Information Displayed

Institution branding, mission overview, service highlights, high-level platform capabilities.

Security Considerations

No PII/PHI displayed; publicly accessible content only.

Data Requirements

Static/CMS-managed marketing content; no patient data.

UX Expectations

Dark hero section with clear value proposition; light-theme content sections below; smooth scroll transitions; mobile-responsive layout.

7.2 About Us Page

Purpose

Communicate institutional background, mission, and credibility.

Primary Users

Public visitors, prospective patients, partners.

Key Capabilities

Static informational content; optional leadership/team overview.

Information Displayed

Institutional history, mission/vision, accreditation or credibility indicators.

Security Considerations

No PII/PHI; publicly accessible.

Data Requirements

Static/CMS-managed content.

UX Expectations

Consistent typography and light theme; readable long-form content sections.

7.3 Contact Us Page

Purpose

Provide a channel for inquiries and support requests.

Primary Users

Public visitors, patients, prospective patients.

Key Capabilities

Contact form submission; institutional contact details; location/hours information.

Information Displayed

Contact form fields, general institutional contact information.

Security Considerations

Form submissions validated and rate-limited; no sensitive health data collected via this form.

Data Requirements

Inquiry records (non-clinical) routed to appropriate staff.

UX Expectations

Simple, accessible form; clear confirmation state after submission.

7.4 Services Page

Purpose

Describe healthcare services and platform capabilities offered by the institution.

Primary Users

Public visitors, prospective patients, referring providers.

Key Capabilities

Service category listings; descriptions of clinical and platform capabilities.

Information Displayed

Service descriptions; department overviews.

Security Considerations

No PII/PHI; publicly accessible.

Data Requirements

Static/CMS-managed content.

UX Expectations

Organized categories with clear visual hierarchy; interactive expandable sections where helpful.

7.5 Patient Data Page

Purpose

Provide authorized users a secure view of individual patient information.

Primary Users

Physicians, Nurses, Clinical Staff, Admin (role-scoped); Patients (own record).

Key Capabilities

View/update patient profile, diagnosis, medication, history, monitoring, severity indicators (per role).

Information Displayed

PII, PHI, diagnosis, medication, history, next-of-kin, monitoring/severity data.

Security Considerations

Strict RBAC; full audit logging of access and changes; session timeout enforced.

Data Requirements

Patient record data model; real-time monitoring feed where applicable.

UX Expectations

Clear information hierarchy; sensitive-data warnings; minimal-click access to critical status information.

7.6 Dashboard Page

Purpose

Provide role-appropriate operational and clinical visualizations.

Primary Users

Physicians, Nurses, Admin, Executives, Analysts.

Key Capabilities

Configurable widgets/metrics; drill-down into underlying data; export where permitted.

Information Displayed

Operational and clinical metrics per Section 5.4 (aggregate and/or patient-level per role).

Security Considerations

Role-scoped data visibility; aggregate/de-identified views for broader analytics roles.

Data Requirements

Aggregated analytics data sourced from operational and clinical records.

UX Expectations

Clean data-visualization layout; responsive charts; light theme with accent color coding for status/severity.

7.7 Admin Login Page

Purpose

Provide secure authenticated entry point for administrative and clinical staff.

Primary Users

Administrators, Co-Administrators, System Administrators, Clinical Staff.

Key Capabilities

Multi-factor authentication; password reset workflow; account lockout after failed attempts.

Information Displayed

Login credentials only; no clinical data displayed.

Security Considerations

MFA required; brute-force protection; secure session handling.

Data Requirements

Authentication records; login attempt logs.

UX Expectations

Minimal, focused, dark-hero login layout consistent with brand; clear error messaging without leaking account existence.

7.8 AI Features Page

Purpose

Provide access to governed AI capabilities (educational content, narration, AI-assisted reporting request, image/audio generation).

Primary Users

Patients (education/accessibility features); Admin/Analysts (AI-assisted reporting); AI Governance Personnel (review).

Key Capabilities

Request AI-generated educational content; request narration/audio; request AI-assisted report draft; view AI review/approval status.

Information Displayed

AI request inputs/outputs; review/approval status; Responsible AI labeling.

Security Considerations

All AI-generated patient-facing/clinical-support content subject to governance review before release; full logging of AI requests and outputs.

Data Requirements

AI request/response records; governance review status records.

UX Expectations

Clear AI-generated labeling; transparent status indicators for content pending review; accessible narration controls.

7.9 Reporting Page

Purpose

Provide access to operational, clinical, and administrative reports.

Primary Users

Admin, Analysts, Executives, Compliance Personnel.

Key Capabilities

Generate, view, and export standard reports; request AI-assisted report drafts.

Information Displayed

Operational/clinical/administrative metrics and report metadata.

Security Considerations

Role-scoped report access; audit logging of report generation and export.

Data Requirements

Aggregated and structured report data sourced from operational/clinical records.

UX Expectations

Structured report layout with filter/export controls; clear distinction between finalized and AI-drafted reports.

7.10 Admin Portal

Purpose

Central workspace for administrative account, role, and permission management.

Primary Users

Administrators, System Administrators.

Key Capabilities

Create/manage administrator and co-administrator accounts; assign roles/permissions; manage approval workflows.

Information Displayed

Administrative account data, role/permission configuration, approval workflow status.

Security Considerations

Highest-privilege access; MFA required; full audit logging of all administrative actions.

Data Requirements

Administrative account and configuration records.

UX Expectations

Structured, table-driven interface with clear confirmation steps for sensitive actions.

7.11 Admin Dashboard

Purpose

Provide administrators with institution-wide operational and governance visibility.

Primary Users

Administrators, System Administrators, Executives.

Key Capabilities

View operational KPIs, audit summaries, AI governance status, system health indicators.

Information Displayed

Institution-wide operational metrics, audit summary data, AI governance review status.

Security Considerations

Restricted to administrative roles; aggregate/summary data emphasized over individual patient detail.

Data Requirements

Aggregated operational, audit, and governance data.

UX Expectations

Executive-style dashboard layout; light theme with clear visual hierarchy and status color coding.

7.12 Site-Wide Design Standards

- Modern, responsive design adapting cleanly across desktop, tablet, and mobile breakpoints.
- Light primary theme for content areas with dark hero sections for key landing/entry points.
- Contemporary/futuristic typography paired with strong readability for clinical and administrative content.
- Purposeful, non-distracting interactive elements and scroll effects that support comprehension rather than decoration.
- Consistent status/severity color coding used across dashboards and patient data views.
- Accessibility-first component design (contrast, keyboard navigation, screen-reader and narration compatibility).

8. Success Criteria

Success is evaluated across seven dimensions, each with representative criteria the delivered platform should satisfy.

8.1 Healthcare Success Criteria

- Clinical staff can retrieve complete patient history, diagnosis, and medication data from a single interface.
- Patient monitoring and severity indicators are visible to authorized staff within defined refresh intervals.
- Admission and discharge workflows are fully documented and auditable.

8.2 Business Success Criteria

- Defined business objectives in Section 2 are measurably progressing against their KPIs post-launch.
- Administrative and operational processing times improve relative to pre-platform baseline.
- Platform adoption meets defined target usage rates across stakeholder groups.

8.3 Administrative Success Criteria

- Administrators and co-administrators can fully manage accounts, roles, and permissions within the platform.
- All sensitive administrative actions produce a complete, retrievable audit trail.
- Approval workflows function as designed with no unauthorized bypass paths.

8.4 Security Success Criteria

- No unresolved critical or high-severity security findings at go-live.
- 100% of PII/PHI access is logged, attributable, and restricted by role.
- Multi-factor authentication enforced for all administrative and clinical accounts.

8.5 Responsible AI Success Criteria

- All AI-generated patient-facing and clinical-support content passes governance review before release.
- AI-generated content is clearly and consistently labeled throughout the platform.
- Responsible AI governance framework is documented, approved, and operational at launch.

8.6 Analytics Success Criteria

- Core operational and clinical metrics defined in Section 5.4 are available on role-appropriate dashboards.
- Metric data reconciles accurately with underlying operational and clinical records.
- Dashboards are actively used by their target stakeholder groups post-launch.

8.7 User Experience Success Criteria

- Core workflows meet defined task-success and satisfaction benchmarks in usability testing.
- Site-wide design standards (Section 7.12) are consistently applied across all pages.
- Accessibility conformance target (e.g., WCAG 2.1 AA) is met across patient-facing and clinical interfaces.

Document Approval

This volume is submitted for executive review and sign-off by the stakeholders below prior to advancement into detailed design and engineering.

ROLE	NAME	SIGNATURE	DATE
Executive Sponsor			
Chief Information Officer			
Chief Technology Officer			
Chief Information Security Officer			

This concludes Volume I of the Business Requirements Document. Application source code is out of scope for this volume; subsequent volumes address security and governance, enterprise architecture, and data, analytics, and design specifications.